

**A Deeper Analysis of the Factors That Inform Matching on Dating Apps:
The Relative Contribution of Pictures, Descriptions and Individual Differences**

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Abstract

The use of dating apps has risen dramatically in the past decade. This change from face-to-face contact to quick-paced decisions on curated profiles has implications for the details individuals focus on when approaching partner selection. Authors De La Mare & Lee (2023) examined the effect of personality compatibility on the likelihood to match with a profile. They found that participants tended to match on the Big Five dimensions of personality with profiles they matched with. In this study, we re-analyzed their dataset to investigate the degree to which profile pictures, descriptions and individual differences contributed to the match decision. We predicted that profile pictures would affect the match decision (RQ1), and have a larger effect than descriptions on this decision (RQ2), as pictures formed the first impression on these profiles. We found that pictures, descriptions, and individual differences all contribute to the match decision. Further, individual differences had the largest variance component, followed by description and then pictures. In the discussion, we explain the implications of these results and the limitations of the dataset.

A Deeper Analysis of the Factors That Inform Matching on Dating Apps: The Relative Contribution of Pictures, Descriptions and Individual Differences

The rapid expansion of online dating platforms has fundamentally transformed the process of romantic partner selection. Whereas romantic relationships were traditionally initiated through existing social networks, educational institutions, workplaces, or geographical proximity, digital dating platforms now provide unprecedented access to large numbers of potential partners. As a result, online dating has evolved from a relatively niche activity into a mainstream mechanism of relationship formation. Finkel et al. (2012) identified online dating as one of the most consequential developments in contemporary relationship formation, arguing that it fundamentally alters how individuals encounter and evaluate potential partners. More recent reviews confirm the continued growth and societal significance of online dating, highlighting its central role in shaping modern romantic experiences across diverse populations (Thomas et al., 2025; Diesen et al., 2025).

Despite providing access to an unprecedented number of potential partners, online dating simultaneously creates a unique decision-making problem. Users are required to evaluate strangers based on highly restricted information, often before any direct interpersonal interaction has occurred. Finkel et al. (2012) argue that this reverses the traditional sequence of relationship development. In offline settings, impressions typically emerge gradually through repeated interactions and social observation. In contrast, online dating users must first decide whether a potential partner appears desirable and only subsequently determine whether genuine compatibility exists. Consequently, online dating requires individuals to make socially consequential decisions under conditions of substantial informational uncertainty.

This challenge is particularly important because partner selection is neither random nor arbitrary. Decades of research demonstrate that individuals systematically evaluate potential partners on a wide range of characteristics, including physical attractiveness,

personality traits, values, lifestyle preferences, and perceived compatibility. Evolutionary perspectives suggest that mate preferences have been shaped by recurrent adaptive challenges, leading individuals to attend to characteristics that may signal partner quality or relationship suitability (Buss & Schmitt, 1993). Similarly, research on assortative mating consistently demonstrates that romantic partners resemble one another on numerous demographic, educational, behavioral, and psychological dimensions (Harper & Zietsch, 2025). Gonggrijp et al. (2023), for example, show that partner similarity is not solely the consequence of shared social environments but also reflects active preferences for particular partner characteristics. Together, these findings suggest that individuals are selective when evaluating potential romantic partners and actively use available information to infer compatibility.

The informational challenges associated with online dating are further complicated by the diversity of motivations that bring individuals to these platforms. Although online dating is often associated with the search for long-term romantic relationships, users pursue a broad range of goals, including casual relationships, sexual encounters, companionship, social validation, and exploration of personal identity. McPherson et al. (2025) demonstrate that online dating motivations are highly heterogeneous and vary across demographic groups and relationship contexts. Consequently, profile information serves multiple functions simultaneously: it not only communicates who a person is but also signals what kind of relationship they may be seeking. This increases the importance of profile construction and interpretation within digital dating environments.

Rather than being neutral descriptions of the self, profiles are carefully constructed self-presentations designed to influence how others perceive and evaluate potential partners. This observation is consistent with broader theories of impression management and computer-mediated communication. Walther's (1996) hyperpersonal model proposes that online environments provide users with increased control over self-presentation, enabling them to

selectively communicate desirable characteristics while minimizing less favorable aspects of the self. Compared to face-to-face interactions, online communication allows greater opportunities for message editing, strategic disclosure, and impression management.

Research on online dating consistently supports this perspective. Ellison et al. (2006) found that online daters continuously negotiate a tension between authenticity and self-enhancement. Participants sought to present themselves in an attractive manner while simultaneously maintaining sufficient accuracy to avoid future disappointment when meeting potential partners offline. Similarly, Toma & Hancock (2010) demonstrate that online dating profiles frequently involve selective self-presentation, with users strategically managing information related to attractiveness and desirability. More recently, Zhang et al. (2025) showed that profile construction involves deliberate choices regarding photographs, textual descriptions, and personal disclosures, all aimed at producing favorable impressions among potential partners. Collectively, this literature suggests that dating profiles should be understood as strategic signals rather than objective representations of identity.

Within these profiles, two forms of information are particularly prominent: profile photographs and written self-descriptions. Both communicate relevant information, yet they do so through fundamentally different mechanisms. Photographs provide immediate visual cues regarding appearance, age, health, style, and perceived attractiveness. Written descriptions, in contrast, communicate less visible characteristics such as personality, values, interests, goals, and lifestyle preferences. Understanding how users integrate these distinct information sources is therefore essential for understanding decision-making in online dating environments.

There are several reasons to expect profile photographs to exert a particularly strong influence on matching decisions. First, physical attractiveness has long been recognized as a central component of romantic attraction and mate selection (Buss & Schmitt, 1993; Toma &

Hancock, 2010). Fiore et al. (2008) argue that physical characteristics often constitute the earliest information considered when evaluating potential romantic partners, making visual information especially salient during initial stages of partner selection. In online dating environments, where users are often exposed to large numbers of profiles in rapid succession, photographs may become even more influential because they provide information that can be processed quickly and efficiently.

Insights from social cognition research further support this possibility. Willis & Todorov (2006) demonstrated that individuals form impressions of attractiveness, trustworthiness, competence, likeability, and aggressiveness after as little as 100 milliseconds of exposure to an unfamiliar face. Remarkably, these impressions closely resemble judgments formed under conditions without time constraints, suggesting that first impressions emerge almost instantaneously. Such findings indicate that visual evaluations operate through rapid cognitive processes requiring minimal deliberation. Given that many online dating decisions involve quick evaluations of unfamiliar individuals, these mechanisms may substantially shape matching behavior.

Evidence from the literature on thin-slice judgments points in the same direction. Ambady and Rosenthal (1992) demonstrated that observers can form surprisingly accurate social judgments from extremely brief observations of behavior. Across a wide range of contexts, evaluations based on limited information often predicted outcomes with notable accuracy. Together, the findings of Ambady and Rosenthal (1992) and Willis and Todorov (2006) suggest that individuals are capable of extracting meaningful social information from minimal visual input. Consequently, profile photographs may function as highly influential sources of information during online dating evaluations.

Nevertheless, a focus on visual information alone would provide an incomplete account of partner selection. Written profile descriptions offer access to dimensions of a

potential partner that cannot easily be inferred from physical appearance. These descriptions communicate personality characteristics, interests, values, attitudes, and relationship goals that may be directly relevant to assessments of compatibility. Indeed, several lines of research suggest that such information plays an important role in romantic decision-making.

Research on assortative mating indicates that individuals often seek partners who resemble themselves on meaningful characteristics (Harper & Zietsch, 2025). Gonggrijp et al. (2023) further demonstrate that active preferences contribute substantially to partner similarity, implying that individuals use available information to evaluate compatibility rather than relying exclusively on physical attraction. Similarly, De La Mare and Lee (2023) found evidence that online dating users exhibit preferences for personality characteristics that resemble their own, particularly with respect to agreeableness, openness, and extraversion. Such findings suggest that textual information communicating personality traits may influence attraction independently of physical appearance.

Moreover, contemporary research on online self-presentation indicates that users intentionally employ written descriptions to communicate desirable personal characteristics. Zhang et al. (2025) found that profile descriptions frequently serve as mechanisms through which individuals communicate identity, values, lifestyle preferences, and social aspirations. Consequently, textual information may provide users with opportunities to evaluate dimensions of compatibility that are inaccessible through visual information alone. Rather than functioning as competing sources of information, photographs and descriptions may therefore provide complementary signals that jointly shape romantic decision-making.

Despite extensive research on online dating, self-presentation, attraction, personality preferences, and partner selection, surprisingly little is known about the relative contribution of visual and textual information to online dating decisions. Existing research has established that both physical attractiveness and personality-related characteristics influence romantic

evaluations. However, these characteristics are typically presented simultaneously within real-world dating profiles, making it difficult to estimate their independent effects. As a result, the relative importance of profile photographs and written descriptions remains poorly understood.

The present study addresses this gap by re-analysing a unique experimental dataset originally collected by De La Mare and Lee (2023). A distinctive feature of this dataset is that profile photographs and profile descriptions were randomly combined rather than originating from the same individual. This design introduces independent variation in visual and textual information, creating a rare opportunity to disentangle their respective contributions to online dating decisions.

The central research question of the present study is therefore:

“To what extent do profile images and profile descriptions contribute to online dating match decisions, and which of these information sources explains a greater proportion of variance in matching behavior?”

Based on previous research highlighting the importance of physical attractiveness, rapid impression formation, and visual information processing in romantic evaluations, profile images may be expected to explain a larger proportion of variance in matching decisions than profile descriptions. At the same time, substantial evidence indicates that textual information communicates personality-related characteristics and compatibility cues that are also relevant for partner selection. Consequently, the present study adopts a cautious and largely exploratory perspective regarding the relative magnitude of these effects.

Original Study

The data analyzed in the present project originate from the study “Assortative Preferences for Personality and Online Dating Apps: Individuals Prefer Profiles Similar to Themselves on Agreeableness, Openness, and Extraversion” conducted by De La Mare and

Lee (2023). The original study was motivated by a longstanding question within relationship science: why do romantic partners tend to resemble one another on a variety of psychological characteristics?

Previous research has consistently demonstrated that romantic partners are often similar in terms of educational attainment, values, attitudes, personality traits, and other individual characteristics (Gonggrijp et al., 2023; Harper & Zietsch, 2025). One possible explanation is assortative mating, the tendency for individuals to prefer partners who resemble themselves on personally relevant characteristics. However, observed similarity between partners does not necessarily imply active preference. Similarity may also arise through social homogamy, shared environments, or gradual convergence over the course of a relationship. Consequently, establishing whether individuals actively prefer similar partners requires a research design capable of isolating preferences before relationships are formed.

To address this issue, De La Mare and Lee (2023) developed an experimental online dating paradigm. The choice of an online dating context was particularly appropriate because digital dating platforms require users to make repeated decisions about potential romantic partners based on limited profile information. Such environments therefore provide a controlled setting in which partner preferences can be studied before longer-term interpersonal dynamics emerge.

Participants first completed the Big Five Inventory (BFI-44), providing measures of the five major personality dimensions: openness, conscientiousness, extraversion, agreeableness, and emotional stability. Subsequently, participants evaluated a series of dating profiles and indicated whether they would be interested in matching with each profile. The dependent variable was therefore a binary match decision, closely resembling the accept-or-reject judgments commonly observed on contemporary dating platforms.

A central methodological innovation of the study was the construction of the dating profiles themselves. Each profile consisted of a facial photograph and a written self-description designed to communicate personality-related information. Importantly, photographs and descriptions were randomly paired rather than originating from the same individual. This procedure reduced potential confounding between visual appearance and personality information while simultaneously creating a highly controlled experimental environment. As a result, participants evaluated profiles that varied independently in visual and textual content.

The original analyses focused on whether participants displayed assortative preferences for personality traits. Using mixed-effects logistic regression models, De La Mare and Lee (2023) examined whether participants were more likely to match with profiles whose personality characteristics resembled their own. The results indicated evidence of assortative preferences for agreeableness, openness, and extraversion, suggesting that individuals preferred potential partners who shared these personality characteristics. In contrast, no significant evidence was found for assortative preferences on conscientiousness or emotional stability.

Although the original study was designed to investigate personality-based assortative preferences, the structure of the dataset makes it valuable for addressing a broader range of research questions. In particular, the random pairing of photographs and profile descriptions creates a unique opportunity to examine how different forms of profile information contribute to online dating decisions. This methodological feature forms the basis of the present study.

Present Research

The present study builds upon the experimental framework developed by De La Mare and Lee (2023) but addresses a different theoretical question. Whereas the original study

focused on assortative preferences for personality traits, our interest lies in understanding how individuals use different forms of profile information when making online dating decisions.

This question emerges from two complementary strands of literature. On the one hand, research on attraction, physical appearance, and first impressions suggests that visual information may exert a disproportionate influence on romantic evaluations. Physical attractiveness is consistently identified as an important component of mate selection (Buss & Schmitt, 1993; Toma & Hancock, 2010), while research on thin-slice judgments demonstrates that individuals can form meaningful impressions based on extremely limited visual information (Ambady & Rosenthal, 1992). Similarly, Willis and Todorov (2006) showed that impressions based on facial appearance emerge within fractions of a second and remain remarkably stable even when additional viewing time is provided. Taken together, this literature suggests that profile photographs may play a dominant role during the initial stages of online dating evaluation.

On the other hand, research on self-presentation and partner preferences highlights the importance of information extending beyond physical appearance. Written profile descriptions communicate personality traits, values, interests, and lifestyle characteristics that may be relevant for assessing interpersonal compatibility. Studies of online dating profiles indicate that users devote considerable effort to constructing textual descriptions that communicate desired aspects of identity (Ellison et al., 2006; Zhang et al., 2025). Furthermore, evidence from personality-based matching research suggests that such information may meaningfully influence attraction and partner preferences (De La Mare & Lee, 2023).

Despite extensive research on both visual attractiveness and personality-related preferences, relatively little is known about the extent to which these information sources independently contribute to matching behavior. This gap persists largely because photographs

and descriptions are typically inseparable in real-world dating profiles. Consequently, determining their relative influence is methodologically challenging.

The dataset used in the present study provides a rare opportunity to address this limitation. Because photographs and descriptions were randomly combined across profiles, the influence of visual and textual information can be examined separately. Rather than asking whether individuals prefer partners with similar personalities, the present study asks a more fundamental question about online dating decision-making itself: which information source contributes more strongly to users' willingness to match with a potential partner?

Answering this question contributes to a growing literature on online partner selection by clarifying the relative importance of visual and textual profile information. More broadly, it provides insight into how individuals navigate information-rich but socially constrained environments when making romantic decisions.

Methods

Expectations

The current study aimed to examine the relative contribution of profile photographs and descriptions to self-rated likelihood to match (match rating). In the original study, the central focus was the formation of a favorable first impression based on the personality compatibility between the participants themselves and a hypothetical dating partner. However, each description came with a randomly selected photograph of either a woman or a man (depending on the reported sexual orientation). We expected that the profile photograph would affect match rating (RQ1) and explain a larger or similar portion of the variance (RQ2).

Participants

A total of 836 participants filled in the matchmaking survey. Participants who did not express interest in the opposite sex, didn't fill in the whole survey or reported being

transgender or non-binary were excluded from the analysis. After this pre-analysis, 589 participants remained in the final sample. Roughly 53% of these participants were women ($n = 312$), the remaining 47% were men ($n = 280$). The majority, around 80%, identified as heterosexual ($n = 471$) whereas 20% was not exclusively heterosexual but expressed attraction to the opposite sex ($n = 113$). Participants in our sample were single (53%; $n = 315$) slightly more frequently than in a relationship (44%; $n = 262$), and a minority reported another option (3%; $n = 15$). Experience with dating apps was split into yes (58%; $n = 340$), no (41%; $n = 262$) and another option (1%, $n = 9$). The mean age was 25.

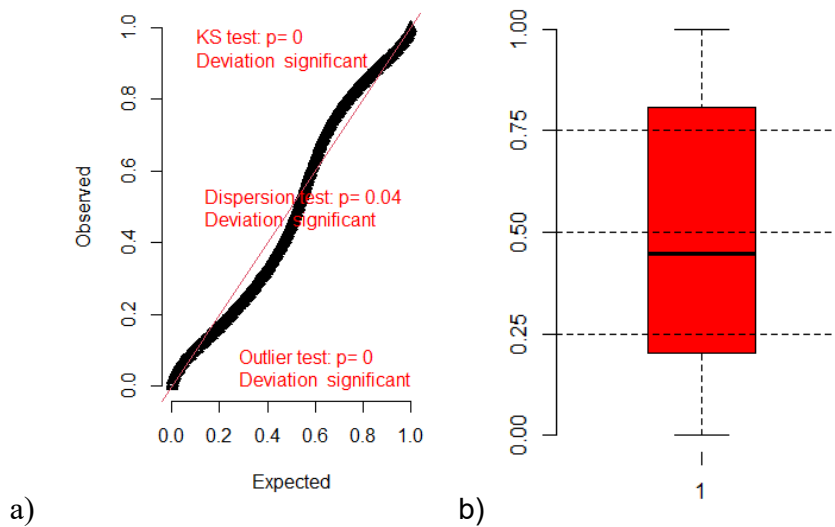
Analysis

To examine the relative contribution of the descriptions and pictures, the original dataset was restructured such that each column only contained one trial. Each unique Description and picture was given an identifier that could be recognized across participants. For every trial, the participant's ID, profile photograph and description were identified and stored as separate variables.

A normality test was performed on this data. Due to the high number of observations, some deviations were found in the normality. However, upon visual inspection and considering the sample size, these deviations were not considered alarming.

Figure 1

Normality QQ-Plot (a) and Simulated Boxplot Based on Residuals (b)



Note. a) A QQ-plot showing the distribution of the residuals. The red line indicated the expected trajectory, whereas the black line indicated the real trajectory. Deviations are considered significant by the listed tests. No pattern was identified in the deviations. b) A box-plot which was simulated based on the distribution of the residuals.

Variance in match ratings was analyzed using a cross-classified linear mixed model, a model type which can be utilized to analyze both fixed and random effects that are either hierarchical or not. To illustrate this difference, assume a particular person only works in one company. The relationship of company to person is thus hierarchical. If we were instead to study this individual employees' contact list, this will likely include people that work in the company and others who do not (i.e. contacts have a non-hierarchical relationship with the company). In this study, the model is non-hierarchical. This results from the factorial design wherein each profile photograph and description can be randomly combined (the pool of pictures which is randomized still depends on the participants' gender and sexuality).

The model was fitted with a restricted maximum likelihood estimation (REML) using the *lme4* package in Rstudio. The match rating was used as the dependent variable. Random

intercepts were used from the participant ID (ID), profile photograph (Picture) and description (Description), resulting in the following formula:

$$\text{Match rating} \sim (1|\text{ID}) + (1|\text{Picture}) + (1|\text{Description}) + (1|\text{Picture:Description}),$$

where Match rating is the independent variable and ID, Picture, Description and the interaction between Picture and Description are random effects. This specifications answers our main research question into the variance attributable to the Picture, Description and individual differences. Additionally, it allows us to control for a potential interaction effect between Picture and Description.

To estimate the relative contribution of each variable, variance components were computed based on the fitted model. Further, likelihood-ratio tests were conducted by comparing the full model with nested models in which random effects were systematically removed. These comparisons were performed using maximum likelihood estimation, as recommended for mixed model comparisons. We only kept factors that were significant at the 0.01 level to correct for multiple comparisons. As a result, the interaction effect was dropped (see next section).

Results

We started our analysis with the full model described by the formula in the previous section. This was compared to a nested model without the interaction effect (1|Picture:Description). There was no significant difference at the 0.01 level as revealed by our likelihood ratio tests ($\chi^2(1) = 4.0119$, $p = 0.045$). This indicates that the descriptions were not consistently interpreted differently depending on the facial features or attractiveness of the picture. Likewise, this implies a process where the attractiveness of a picture was determined separately from the description. Second, we compared the three factor model (Match rating $\sim (1|\text{ID}) + (1|\text{Picture}) + (1|\text{Description})$) to a two-factor model with only ID and Description. Picture contributed significantly ($\chi^2(1) = 1292.5$, $p < .001$) to the variance. Description ($\chi^2(1)$

= 3239.2, $p < .001$) and participant ID ($\chi^2(1) = 9198.8$, $p < .001$) likewise contribute significantly to the full model. We determined that all three factors were important in the match-making decision. This model is congruent with RQ1.

We calculated the proportion of variance explained by ID, Picture and Description using a linear mixed effects model. Participant-level differences explained the highest level of variance at 19.2% ($\sigma^2 = 0.047$). These results show that the biggest factor in our model was the individual differences between participants in how highly they rated the profiles overall. This large variance component could be explained by a different selection strategy. Certain people are more selective and may only match with highly attractive options, whereas others cast a wide net of potential romantic options. Picture and Description accounted for 2.8% ($\sigma^2 = 0.007$) and 5.7% ($\sigma^2 = 0.014$) respectively. The descriptions contributed roughly twice as much to the overall variance as the pictures. This contradicts RQ2 that pictures inform most of the match-making decision. There are multiple possible explanations for this finding. The first is that the participants were more focused on the descriptions than the pictures, since they had just taken a personality test. Secondly, they were in a laboratory setting which could have influenced them to display socially desirable behavior, for example wanting to seem objective and open-minded. Thirdly, the pictures were neutral, front-facing photographs and were perhaps not sufficiently informative or attractive for participants to prioritize them over text descriptions. In a real dating app scenario, profiles tend to contain several pictures and can themselves inform about a person's interests or general personality.

The remaining 72.4% of variance was residual. This is a logical result, as the main research goal of the original study was personality compatibility, and this was not considered in our model. This decision was made because the original personality survey data was no longer available. This variance distribution was confirmed by an intraclass correlation test ($icc = 0.276$).

Discussion

One of the biggest challenges in this study was the lack of transparency regarding the coding of the original data provided online by De La Mare and Lee. The absence of a key detailing the exact meaning of the variable coding created the necessity to deduce the meaning of encoded answers from the context (e.g. combinations with other answers). For starters, the variable sexual orientation assumed 5 possible outcomes, noted using numbers 1 to 5. Which sexual orientations corresponded to each number was not mentioned in the original paper or supplementary materials. However, the original study did state that “participants who did not identify as cis-gendered ($n = 39$), did not report being attracted to the opposite sex ($n = 99$)” (De La Mare & Lee, 2023) were removed from the data, as the research only aimed to investigate heterosexual mate preferences. This meant excluding a sizable group of about 250 participants from the original sample of 836. The R-code in the supplementary materials provided by the researchers, showed that their analysis included participants who with sexual orientation 1 or 3. For the sake of this study, we assumed 1 corresponds with heterosexuality (as it is the most common kept in the dataset and an intuitive choice), 2 corresponds with homosexuality and 3 with bisexuality. The next step was then to remove all participants whose given gender did not match that of their given sex, as the original study only concerned cis-gender people. Since both sex and gender were coded as 1 and 2, it can be deduced that 1 is male as participants who gave this answer were given pictures of women to rate (this deduction requires that our former assumption of number 1 corresponding with heterosexuality is true). More confusion stemmed from the fact that some of the remaining participants rated pictures of the same sex, despite having stated their heterosexual orientation, with no mention of this being a deliberate decision in the original study. Since these answers were not ratings of the opposite gender by a heterosexual individual, these data points were also removed from the analysis. Finally, like De La Mare

and Lee did, all participants with incomplete data - be it on the metadata, personality test or test phase - were removed.

As expected, significant results for the effect of photographs on profile judgement were detected even at the 0.01% significance level, supporting RQ1 as well as previous studies (Buss & Schmitt, 1993; Fiore et al., 2008; Toma & Hancock, 2010) which found that physical attraction plays a large role in the selection of romantic partners. Contrary to RQ2, results suggested that a description does explain more of the variance than the photograph, although their individual proportions are just 5.7% and 2.8% respectively, which means that description's explained variance proportion is twice as large as that of the photographs. Despite all this, individual differences were found to be a much stronger predictor for judgements, with 19.2%.

Given a literature that strongly implies the dominance of visual stimuli, we predicted that in a dating-app simulation, participants would be more strongly influenced by photographs rather than the written descriptions. There are several explanations as to why this discrepancy between hypothesis and findings exist. The first is that the original experiment from which the data was collected, started with a short personality test. This test could have steered participants into a personality-based thinking. Another factor could simply be that the visual stimuli were very simple, neutral and standardized photographs without much personality (see figure 2 below). This is contrary to dating apps, in which users are more inclined to upload self-enhancing pictures with the motivation to make themselves look more desirable (Toma & Hancock, 2010). This fact might make photographs seem less convincing in contrast. A third explanation is that due to a lab-setting, participants may be more inclined to give more objective answers, be it as a result of social desirability and not wanting to come across as too superficial, or a desire to help the researchers by deeply considering all of the information that comes with each trial.

Figure 2

Examples of profile photographs of women, to be rated by male participants.



These explanations can suggest two things. The first is that the methodology used by De La Mare & Lee may not sufficiently be ecologically valid to elicit natural dating app behavior for the researchers to observe. The other suggestion is that by using this design of neutral, standardized pictures, you have created a controlled environment in which you are more free to study the written descriptions. Future studies aiming to compare different description styles, could make use of this paradigm and its visual stimuli to control for photograph influence.

The theory of assortative mating might also explain why individual differences was such a strong predictor of whether a person wanted to match with a profile, compared to photographs and descriptions. Standardized photographs meant that participants had to spend more time studying the descriptions. These descriptions can reveal more about the fictional person's personality than a photo would, therefore it might result in an increased chance of approval if a participant's personality resonates with the description. This means that the results of this study can be seen as support for the idea of written descriptions becoming more valuable once visual stimuli have been controlled for. Moreover, this highlights an issue in the original study where written descriptions may have had a much stronger influence on partner selection compared to dating apps where photographs also display a more expressive, idealized self.

AI Acknowledgement

- ChatGPT was utilized to suggest extra details to report in the method section, and to help find ways to create a graph in Rstudio.
- Google's NotebookLM was used to retrieve previously discovered information from papers. An example command would be "Which paper supported the theory of assortative mating?", after which the source of each AI-answer was confirmed by the researcher personally.
- For the introduction section, artificial intelligence tools (ChatGPT, OpenAI) were used for spelling and grammar checking and to support the preliminary identification of relevant literature. The author remains responsible for the final content of this section.

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